



Fast Healthcare Interoperability Resource Genomics and Polygenic Risk Scores

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KEYWORDS

• HL7 FHIR • CDS • Polygenic risk scoring • Whole genome sequencing • Precision medicine

KEY POINTS

- Features of reporting polygenic risk scoring for clinical use.
- Health Level 7's (HL7) Fast Healthcare Interoperability Resource (FHIR) can deliver the polygenic risk scoring features computationally.
- Example of polygenic risk scoring in FHIR.
- A critical feature is the genetic similarity between a patient and the underlying population used to develop the score.
- As the clinical expression of polygenic risk scoring evolves, the FHIR representation can evolve.

INTRODUCTION

Polygenic Risk Scoring

While the American College of Medical Geneticists (ACMG) has recently recommended caution [1,2], polygenic risk scoring is becoming a feature in genomic clinical diagnostic reporting. Standardizing the exchange of critical data elements will aid consistency and accessibility in numerous computational applications, such as clinical decision support (CDS), quality control, end-user user interfaces, searching, analysis, and potentially artificial intelligence. Recognizing the importance of polygenic risk scoring, the ACMG has recently published 3 excellent viewpoints on polygenic risk scores (PRSs) [1–3], which, along with other experts [4–10], aid in understanding the crucial elements and offer vital insights. Several features, such as assessing the study population's and the patient's genetic similarity, are indicated. As well as structuring data associated with polygenic risk in

a consistent exchange format enable dynamic genomic test reporting paradigms to make test results more useable. Indeed, Brockman and colleagues [4] found in user experience testing that “static genomic test reports that limit one's ability to explore results are not ideal for polygenic risk disclosure.” Computable representation lends itself to access through web tools, interactivity, and CDS.

Health Level 7 International

Health Level 7 International (HL7) is an American National Standards Institute (ANSI)-accredited, consensus-driven, nonprofit standards body specializing in computable, standard health care data exchange and retrieval to streamline health services and clinical care [11]. Understanding what being an ANSI standard body means gives insight into HL7. Accredited ANSI standard bodies meet several criteria [12]. The requirements include open-recorded meetings, a consensus-driven

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approach with a balanced interest among large and small partners, following due process/governance, and continual standard(s) improvement. These features are part of HL7 being a community-driven organization that serves the needs of the national and international health care industry, as well as public and private interests. HL7 membership is drawn from across the globe and health care stakeholders. Members include physicians, nurses, community-based care organizations, nongovernmental organizations, governmental organizations, insurance companies, patient advocates, software engineers, solution architects, data scientists, c-suite leaders, and patients. A significant goal of HL7 members is to devise standards that will streamline and support the ability to interoperate, including across national boundaries.

Interoperability is the ability to share and use data between multiple sources [13]. When data are broken down into discrete pieces, it enables computational systems to share information more explicitly [14]. Consider sending laboratory information as discrete bits of data versus a single picture of the result [15]. The discrete data are more useable by 2 computational systems, allowing them to interoperate. In addition, it is useful to standardize the exchange. This reduces the complexity of the solutions required. Consider that an organization such as a hospital interacts with multiple vendors (perhaps several diagnostic laboratories and community-based organizations), creating unique interfaces for each interaction would require repeating the same resource investment, whereas standardizing the interface code base for the exchange of data between systems can have cost savings [16,17]. HL7 has a unique role in establishing standard specifications of how data should be structured in health care data exchange—reducing the landscape of possible choices implementers/developers might make—aiding in consistency and the ability to repurpose or interoperate [14–19]. Interoperability is believed to be so essential that the US Office of the National Coordinator (ONC) has published in HT-1, the final rule, “The adoption of new and revised standards and criteria in this final rule will facilitate interoperability through standardized health information and functionality, which will lead to better care and health outcomes for patients, while reducing burden and costs” [20].

HL7’s Fast Healthcare Interoperability Resource (FHIR) specification has a large implementation and development community [21] (see <https://confluence.hl7.org/display/FHIR>). FHIR is based on the notion of a resource—the smallest unit in FHIR for exchanging information. Resources, such as patient (<https://hl7.org/>

[fhir/R4/patient.html](https://hl7.org/fhir/R4/patient.html)) and organization (<https://hl7.org/fhir/R4/organization.html>), are independent but can cross-reference each other. To be more readily implementable, FHIR takes advantage of modern web standards, such as JavaScript Object Notation (JSON) and Representational State Transfer (REST) architecture. Additionally, the core FHIR Resource specification can be constrained for specific use cases to produce specialized implementation guidance in computable specifications called implementation guides (IGs). Two notable IGs are the US Core (<https://hl7.org/fhir/us/core/>) and Clinical Genomics Reporting (<https://hl7.org/fhir/uv/genomics-reporting/>). To provide an impression of the broad adoption of FHIR, the US ONC, which oversees the electronic health record (EHR) certification process, “adopted the HL7 FHIR US Core Implementation Guide (IG) standard for trial use (STU) version 6.1.0” as the base implementation specification to meet certification requirements for January 2026, this replaces US Core 3.1.1 as the current base. The US Core IG contains US realm use case-specific guidance. For example, to ensure consistent representation of body mass index (BMI) units, the US Core BMI profile (<https://hl7.org/fhir/us/core/STU6.1/StructureDefinition-us-core-bmi.html>) constrains the valueQuantity.code element to code of kg/m². FHIR IGs also provide use case examples and specific narrative implementation guidance to aid implementers. An important facet is that the specification is computable. An IG’s computable artifacts can be directly used for validation or in certification, as seen in Touchstone (<https://touchstone.aegis.net/touchstone/>), Inferno (<https://inferno.healthit.gov/> and <https://inferno.healthit.gov/validator/>), the HL7 FHIR validator (<https://validator.fhir.org/>), or FHIR server implementations (such as <https://learn.microsoft.com/en-us/azure/healthcare-apis/fhir/validation-against-profiles>).

The Clinical Genomics FHIR IG is a universal realm HL7 IG specific to the exchange of genomic data. Universal in this context means the Clinical Genomics IG is intended to be applicable as a base for all international settings and use cases. The Clinical Genomics working group (WG) has published the STU2 version (<https://www.hl7.org/fhir/uv/genomics-reporting/index.html>) of the IG and is preparing to publish STU3. The balloted STU3 version (<https://hl7.org/fhir/uv/genomics-reporting/2024Jan/>) clarifies key areas and adds genomic operations [21]. To help in your implementation efforts or to contribute to the standard, the Clinical Genomics WG weekly call agenda can be found at <https://confluence.hl7.org/display/CGW/CG-2024-03-12>. Also, there is a zulip chat to interface with experts

in FHIR and clinical genomics at <https://chat.fhir.org/#narrow/stream/179197-genomics>, and an HL7 accelerator (<https://confluence.hl7.org/display/COD/Genomics+-+GenomeX>).

The Clinical Genomics IG contains many examples of data instances, computable specifications, and narrative guidance on structuring genetic data in FHIR. The authors explore an example of polygenic risk reporting here. With the specifications of the Clinical Genomics WG and other FHIR core Resources, such as the FHIR RiskAssessment Resource, HL7 FHIR has many tools to meet the growing need to communicate the complete data set of polygenic risk reporting elements in a computable manner that takes advantage of existing paradigms of information exchange.

DISCUSSION

Basic Anatomy of Polygenic Risk Scoring and Risk Reporting

While not an exhaustive summative evaluation, the concepts/data elements that follow were derived from a formative survey of the landscape to collect key features in polygenic risk clinical reporting. The sources vary from ACMG points to consider [1–3], including one focused on laboratory reports [3], to a clinical trial of polygenic risk assessment in clinical practice [5], technical requirements for translating scores into risk [7], to standards around reporting new PRSs [9,22], and usability testing [4,6]. Together, these provide access to the experiences and thoughts of a large and diverse cohort of subject matter experts. Table 1 summarizes the sources.

PRS-derived insights in clinical care are newly evolving, and caution is warranted. Yet, PRS has been used as part of screening tools (eg, [24]) for some time. There is an eagerness to improve patient care with the precision promised by PRS (especially with artificial intelligence’s explosive evolution). Generally, the sources indicate that the risk calculated from a PRS be treated like a screening tool result as expressed in the laboratory ACMG points to consider, “PRS describes a statistical association, and high risk does not indicate an underlying disease pathology ... whereas PRS is solely considered screening information” [3]. This is an essential conceptual shape to remember when considering the data elements for reporting risk assessments derived from PRS.

The features extracted from the landscape survey are presented in Table 2. Note the granularity in the presentation is variable to keep the list reasonable in length (see Table 2).

Clinical Decision Support Standards Used with Fast Healthcare Interoperability Resource: Substitutable Medical Applications, Reusable Technologies on Fast Healthcare Interoperability Resource and Clinical Decision Support Hooks

Clinical decision support systems aid clinical decision-making by providing patient-specific assessments based on individual patient characteristics [27,28]. They have had successes in streamlining aspects of care such as ordering, timing of screening tests, and providing recommendations [27,28]. With regard to FHIR, 2 critical CDS architectural tools are Substitutable Medical Applications, Reusable Technologies (SMART) on FHIR and CDS Hooks.

SMART on FHIR [29] is a tool for launching external applications from an electronic medical record (EMR). This component of the standard relates to app authentication and launching. The specification initially took advantage of the OAuth 2.0 standard to limit the scope of app access to only what was needed. As of 2021, there appeared to be hundreds of SMART on FHIR apps [28], and in the United States, the 2 largest by market share EMRs (as well as most others) support SMART on FHIR applications. Additionally, SMART on FHIR is part of the US ONCs HT-1 final rule [20] as a system that should be supported, along with CDS Hooks. SMART on FHIR apps can vary in complexity of user interaction. Typically, a button launches the app, and a field of view is populated with the information the app is designed to convey. Once launched SMART on FHIR apps can offer all the options for user interaction and exploration seen in modern web browsing.

CDS Hooks [30] uses triggering events in the clinical interaction with EMRs to call Web services. It is RESTful and uses FHIR expressed in JSON to exchange patient data. Since SMART on FHIR apps are also Web services, CDS Hooks can offer the clinician the opportunity to open a SMART on FHIR app contextually or simply launch the SMART on FHIR app. When the app is opened, CDS Hooks feeds context based on the triggering event to the app. For example, suppose a clinician is opening a chart. In that case, a CDS Hooks might call a polygenic risk reassessment app to quietly look for new PRSs from a catalog of scores and calculate whether there are new traits that the patient has an elevated risk for. When the calculation is finished, a “card” will appear with the result. The “card” can have several features, including an indicator of its importance. A low level of importance might cause the card to wait to be opened. Or, if the card is of high importance, it could interrupt

TABLE 1
Source Articles for Data Elements Discussed

Title	Comment
Patient and provider perspectives on polygenic risk scores: implications for clinical reporting and utilization [6]	User viewpoint (patient and provider)
Design and user experience testing of a polygenic score report: a qualitative study of prospective users [4]	User viewpoint (patient)
The clinical application of polygenic risk scores: A points to consider statement of the American College of Medical Genetics and Genomics (ACMG) [2]	Clinical use (ACMG review/statement)
Clinical utility of polygenic risk scores for embryo selection: A points to consider statement of the American College of Medical Genetics and Genomics (ACMG) [1]	Clinical use (ACMG review/statement)
Laboratory perspectives in the development of polygenic risk scores for disease: A points to consider statement of the American College of Medical Genetics and Genomics (ACMG) [3]	Laboratory clinical reporting (ACMG review/statement)
The need for polygenic score reporting standards in evidence-based practice: lipid genetics use case [8]	Sharing of new PRS (translational)
A tool for translating polygenic scores onto the absolute scale using summary statistics [7]	Technical requirements (risk score)
Development of a clinical polygenic risk score assay and reporting workflow [5]	Laboratory clinical reporting (in practice)
Validation of an Abridged Breast Cancer Risk Prediction Model for the General Population [23]	Risk assessment (mixed PRS and other features)
Breast Cancer Screening in the Precision Medicine Era: Risk-based Screening in a Population-based Trial [24]	Risk assessment (mixed PRS and other features/review) [10]
Polygenic risk scores in pharmacogenomics: opportunities and challenges—a mini review [10]	Laboratory clinical reporting/ clinical use (mini review)
Principles in genetic risk assessment [25]	Risk assessment (genetics general) [9]
Improving reporting standards for polygenic scores in risk prediction studies [9]	Sharing of new PRS (translational)—ClinGen and PRS Catalog
Polygenic risk scores in the clinic: Translating risk into action [26]	Laboratory clinical reporting/ clinical use (recommendations)
The Polygenic Score Catalog as an open database for reproducibility and systematic evaluation [22]	Sharing of new PRS (translational)—PRS Catalog

the current clinical workflow. The card itself can contain a link that launches a SMART on FHIR app, perfect for providing a platform to explore a polygenic risk assessment and scoring results.

Fast Healthcare Interoperability Resource Genetics Brief Anatomy

FHIR is a flexible standard. It can be expressed in JSON and XML and includes a robust RESTful API. FHIR can be thought of as a “platform specification” [31]. REST specifies the typical behaviors you see

with Web sites and internet-accessible endpoints. The REST API specifies how to request that an FHIR server search for data. As mentioned, FHIR IGs further adapt the specification to specific use cases. The Clinical Genomics WG has produced and trialed an implementation guide (<https://www.hl7.org/fhir/uv/genomics-reporting/index.html>) for the exchange of genetic data.

The current Clinical Genomic Reporting IG is written on the FHIR core R4 specification. FHIR uses semantic versioning, so the increment from DSTU3 indicates that

TABLE 2
Features

Feature	More Granular Data Element
Treat like a Clinical Screening Test	
Handling Negative or Low-risk Result	
Accessibility for Exploration	
Limitations/Disclaimers	
Description of Test Performed (methodology)	
References and Citations	
Treatment Recommendations	
Graphical Representations	
Represent Risk in Multiple Ways (stating both 90th percentile and top 10%)	
Recommendation for Posttest Counseling	
Impact on Care	
Guidelines	
Educational Materials Accessible for Patients and Provider	
System for Reanalysis	
Variant	Genetic variation used for calculation (including genome build, coordinates, etc...)
Variant	Imputed (measure of confidence)
Variant	Types of variation (indel, single nucleotide polymorphism [SNP], etc...)
Variant	Platform used for observing variation
Trait	Specific outcome defined
Trait	Method of collection (interviews, self-reported, abstracted, or computed from billing codes)
Trait	Compatible tests for Mendelian gene variation linked to the disorder
Trait	Incidence of disease or outcome
PRS (score)	Analytical validity (including performance metrics)
PRS (score)	Clinical validity
PRS (score)	Clinical utility
PRS (score)	Ancestry adjustment
PRS (score)	Other non-PRS attributes used in scoring and relative impact
PRS (score)	Individual weights
PRS (score)	Threshold needed for action with “gray area”
PRS (study population)	Study population PRS mean and standard deviation
PRS (study population)	Characteristics including text description
PRS (study population)	Similarity to a reference population used to represent the patient

(continued on next page)

TABLE 2 (continued)	
Feature	More Granular Data Element
PRS (patient population)	Reference population most similar to the patient (principal component analysis has been indicated here with as many as 5000 variants across the genome)
PRS (patient population)	Characteristics including text description
PRS (patient population)	Patient population PRS mean and standard deviation
PRS (risk)	Continuous representation
PRS (risk)	Dichotomous representation
PRS (risk)	Definition of high-risk status
PRS (risk)	Kind (percentile, relative, or absolute risk)
PRS (risk)	Temporal conditions (eg, over 5 y, at the age of 40 y)
PRS (risk)	Percentile
PRS (risk)	Relative risk
PRS (risk)	Threshold needed for action with “gray area”
PRS (risk)	Temporal conditions (eg, over 5 y at the age of 40 y)
PRS (absolute risk)	Underlying disease prevalence/incidence (unlikely to have models for absolute risk)

breaking changes were made. Currently, HL7 is working on R6. With each increment of the FHIR version, the core specification becomes more “normative.” In HL7 terms, “normative” means backward compatibility with any future specification updates, and more importantly, the HL7 community (made up of participants and stakeholders in the health care field) has determined that the specification is stable and fit for purpose. The Clinical Genomics WG’s IG is on FHIR R4 as that is the maturest version of FHIR to have broad adoption.

The FHIR Resource Observation (<https://hl7.org/fhir/R4/observation.html>) was chosen by the Clinical Genomic WG as the most consistent way to express genetic findings found in laboratory reporting. Observation Resource is a normative Resource, which means it has had wide adoption and is broadly supported. Additionally, the Observation Resource contains a very flexible tool called “component.” The construct of the component is a coded name—value pair. That is, the definition of what data element a particular component represents is conveyed computably within the component through “component.code,” a field that communicates a standard coded concept. For example, the Variant Observation Profile (<https://hl7.org/fhir/uv/genomics-reporting/StructureDefinition-variant.html>) has a component called genomic-source-class (in FHIR speak that would

be Observation.component:genomic-source-class). This component has been assigned LOINC code 48002-0. LOINC is a standard terminology organization specializing in concepts for observations, among other things. The entry for LOINC code 48002-0 (<https://loinc.org/48002-0>) states that the concept is “The genomic class of the specimen being analyzed: Germline for inherited genome, somatic for cancer genome, and prenatal for fetal genome.” Looking at the component Observation.component:genomic-source-class, one can see that the component’s value has been constrained to a value set of terms that answer the question, “What is the genomic source class?” Constraining Resources (or profiles) are how FHIR can provide a more tightly used case-bound specification. The constraints found in the genomic-source class component make it easier to implement in the specific use case of genomic clinical reporting of genetic variation.

The specification structure is a computable artifact, expressed as JSON on <https://hl7.org/fhir/uv/genomics-reporting/StructureDefinition-variant.json>. The code field in the component is the data element where the meaning of the component is conveyed. We already have discussed that it has been given a LOINC code to define it. Looking at the component more closely, one will see that the code is composed of 2 parts: system

and code. The system tells us computably that the code is a LOINC code and that the concept is found wherever <https://loinc.org/> leads. LOINC has a clean URL system that allows the code to be combined with the domain name to retrieve the concept definition.

An essential aspect of identifying the component definition with a codeableConcept is that a server can support searching by component.code concepts. The FHIR REST API (<https://hl7.org/fhir/R4/http.html>) and Search (<https://hl7.org/fhir/R4/http.html#search> and <https://hl7.org/fhir/R4/search.html>) indicate the syntax to perform such a search. The Clinical Genomics Reporting IG has an example query, “: GET/Observation?component-code-value-concept [http://loinc.org|48002-0\\$http://loinc.org|LA6683-2](http://loinc.org|48002-0$http://loinc.org|LA6683-2)” ([https://hl7.org/fhir/uv/genomics-reporting/usecases.html#get-references-to-all-variants-obtained-where-the-analysis-has-b-.](https://hl7.org/fhir/uv/genomics-reporting/usecases.html#get-references-to-all-variants-obtained-where-the-analysis-has-b-)). The “GET” portion indicates using a RESTful GET interaction. This is nothing more than the base URL of an exposed FHIR server endpoint. So, one can think of it as “[https://base-URLofFHIRserver.com/Observation?component-code-value-concept=http://loinc.org|48002-0\\$http://loinc.org|LA6683-2](https://base-URLofFHIRserver.com/Observation?component-code-value-concept=http://loinc.org|48002-0$http://loinc.org|LA6683-2).” The “Observation?” portion indicates that we are performing a search on FHIR Observation Resource data instances. The portion after the “?” is the parameters of the search. In this case, the syntax is using a combined search parameter of component.code and component.valueCodeableConcept (<https://hl7.org/fhir/R4/observation.html#search>). The first part “<http://loinc.org|48002-0>” indicates looking for Observations data instances with a component for genomic-source-class, as defined by component.code having system LOINC and concept code 48002-0. Not all Observations will have the component as the component has cardinality 0 ... 1. This means that there can be 0 to 1 instances of the component.

In addition to components, the Observation Resource has additional fields such as subject (who the observation is about), effective time (clinically relevant period for the observation), and performer (who carried out the observation), a place for a note with attribution (the note field is a coded annotation, see <https://hl7.org/fhir/uv/genomics-reporting/StructureDefinition-coded-annotation.html>). The codedAnnotation data type allows attribution, time stamp, type of note and the note to be captured, method (like sequencing), device, and so on. These are useful for capturing all the data computably that describes an Observation. The Clinical Genomics IG adds several components to the variant so that the observation instance can capture genomic variation. The variant reporting page (<https://hl7.org/fhir/uv/genomics-reporting/sequencing.html>) goes into detail on the

guidance for implementers to communicate genetic variation observations.

Genetic variation is one of the data elements to represent in Polygenic risk scoring, and the HL7 Clinical Genomic WG uses Observation Resource to do that. However, other resources convey the additional data elements. The FHIR Resource RiskAssesment (<https://hl7.org/fhir/R4/riskassessment.html>) is a good fit for the concept of assessed polygenic risk, and the Resource narrative definition states that “list of potential health risks based on a patient’s genetic analysis” is one of the use cases for the Resource. To tie the RiskAssesment to the observed patient genetic variation used in the calculation of risk, the RiskAssesment.basis field is used. RiskAssesment.basis has the data type Reference (<https://hl7.org/fhir/R4/references.html#Reference>) with an indicator that data instances of “Any” Resource type can be referenced. The Reference data type can accommodate different types of referencing, as explained on the specification page, for this discussion, we stop at indicating that the reference serves as a pointer to a different data instance. Additionally, RiskAssesment.basis has cardinality 0...*, meaning multiple FHIR Observation Variant Profiles can be referenced.

Terminology

While FHIR does a very excellent job of providing a standard structure, syntactic standard, FHIR core only partially solves the semantic standardization problem. For example, where coded concepts are communicated, there can be differences between systems. Luckily, the Resource profiling constraints inherently provide a mechanism for semantic clarity. For example, the Variant Profile (<https://hl7.org/fhir/uv/genomics-reporting/StructureDefinition-variant.html>) was constrained in Clinical Genomics Reporting IG. However, in the HL7 FHIR context, it is important that a community determines what terms (coded concepts) will be used and where the definition of those terms will reside. The HL7 Clinical Genomics WG often collaborates with LOINC when new LOINC codes are needed as genomic concepts mature. Additionally, looking at SNOMEDCT, one finds many disease-related concepts. These are often cited in HL7 standards as the coded concepts to use to communicate conditions or problems that a patient might have. The governance and constraining of allowable coded concepts is an open question, but IGs help in constraining to a use-case specific list of concepts, and HL7’s Terminology oversight group maintains a large body of sets of concepts grouped for specific

purposes (value sets, see <https://terminology.hl7.org/>) to aid implementers.

Polygenic Risk Score Reporting Risk example

As an example of expressing PRS concepts in FHIR for clinical reporting, 2 key FHIR core Resource instances were populated: the FHIR Risk Assessment Resource and the FHIR Observation Resource.

Fig. 1 shows the FHIR RiskAssessment instance. In English, it reads, “CGPatientExample01 (subject) is predicted (prediction) to have a high risk (qualitativeRisk) with a relative risk of 2.69 (relativeRisk) for developing Diabetes mellitus type 2 (outcome) at age 65 (whenRange). The basis is seven genetic variants and a polygenic score observation (basis). The RiskAssessment is related to the polygenic score observation (parent). And there is a disclaimer (note with extension).” If used, the field “mitigation” would guide the steps to take.

It should be noted that a genetic file extension could have been used instead of the 7 observed variants. It is an important discussion item to determine how to handle potentially millions of genetic variants. Luckily, FHIR supports retrieval and relative links. There is a whole set of FHIR Genomic Operations that might be brought to bear here with a preconfigured GET request [21].

Fig. 2 is the polygenic score observation, an FHIR Observation instance. The example shows how components can be used to provide additional data elements. For searching, the Observation.code value is a LOINC code for PRS. This helps make the data instance more discoverable and allows a receiving system to develop specific rules for handling PRS data received in FHIR as part of Genomic Reports. The data instance uses the genomic-risk extension to connect the Observation to the RiskAssessment data instance. It is done through a reference so the RiskAssessment can be accessed appropriately. Additionally, the genomic-artifact extension is being used to provide a citation with study details on the development of the assay, including validation, validity metrics, and descriptions of the development of the score (including population used).

Here, we have taken advantage of FHIR Resource Observation’s ability to use “components” to communicate 7 PRS-specific data elements. The components are prs-ancestry-similarityToValidationPopulation, prs-study-dis-mean, prs-study-dis-sd, prs-ancestry-adj-dis-mean, prs-ancestry-adj-dis-sd, prs-ancestry-adj, and prs-percentile. Although these components are not yet in the Clinical Genomics IG and use TBD code, FHIR servers can accept, share, and perform searches on them as they are. However, downstream implementation may lag until the components are widely accepted and part of an

FHIR IG. When the broader medical genomics community has determined the features for PRS reporting, they can be added to the IG, perhaps as a profile, and LOINC codes requested. The process takes 6 to 9 months. It should be stressed that FHIR servers, thanks to the FHIR core specification, can accept, share, and search on them as they are, even before they are part of an IG. The FHIR Observation component is a versatile communication tool.

The component “prs-ancestry-similarityToValidationPopulation” shows that the patient is 95% likely to be from the population found in the study that produced the PRS. This is an important metric as it gives confidence in the PRS’s applicability to the patient’s health. Principal component analysis (PCA) is one tool used when adjusting an individual’s score based on genetic ancestry differences from the study population [5]. It is from the PCA that the measure of likelihood can be determined, as well as a tool to adjust scores to account for ancestry. However, PRS clinical reporting has not reached the state where this is considered standard practice. The similarity measure is an essential determinant in a clinician’s decision on the meaningfulness of the test result for the clinician’s specific patient. In the case of 95%, the score is probably relevant for the patient.

The paired components prs-study-dis-mean and prs-study-dis-sd provide the mean and standard deviation of the distribution of scores for the study population. This allows one to calculate a graphic to show this distribution visually. Also, the paired components prs-ancestry-adj-dis-mean and prs-ancestry-adj-dis-sd provide the means to visualize the adjusted distribution of the population which the patient most matched in the PCA mentioned earlier. These distributions serve as a backdrop to the adjusted score (communicated with the component PRS ancestry adjusted) and the ancestry-adjusted PRS percentile (communicated with the component prs-percentile). The percentile’s impact is better understood with some knowledge of the distribution.

Finally, the Observation example in Fig. 2 shares a disclaimer using the note data element. A disclaimer that appropriately indicates that the score has been adjusted for ancestry and the similarity to the study population is an essential caveat in interpreting the result.

The examples show that FHIR standards tools can be used to computationally communicate key concepts and features needed to report and utilize PRS risk. FHIR also contains data elements that capture important metadata such as performers, patient identifiers, and the like. FHIR core offers robust coverage of most data elements in the clinical space.

```

{"resourceType": "RiskAssessment",
  "id": "GenRiskDiabetesT2atAge65",
  "parent": {"reference": "Observation/polyGenicScoreObservation"},
  "status": "final",
  "subject": {"reference": "Patient/CGPatientExample01"},
  "basis": [
    {"reference": "Observation/polyGenicScoreObservation"},
    {"reference": "Observation/VariantExample1"},
    {"reference": "Observation/VariantExample2"},
    {"reference": "Observation/VariantExample3"},
    {"reference": "Observation/VariantExample4"},
    {"reference": "Observation/VariantExample5"},
    {"reference": "Observation/VariantExample6"},
    {"reference": "Observation/VariantExample7"},
    {"reference": "Observation/VariantExample8"},
    {"reference": "Observation/VariantExample9"}
  ],
  "prediction": [
    {
      "outcome": {
        "coding": [{"code": "44054006", "system":
"http://snomed.info/sct", "display": "Diabetes mellitus type 2
(disorder)"}]
      },
      "qualitativeRisk": {
        "coding": [{"code": "high", "system":
"http://terminology.hl7.org/CodeSystem/risk-probability", "display": "High
likelihood"}]
      },
      "relativeRisk": 2.69,
      "whenRange": {"high": {"value": 65, "code": "a", "system":
"http://unitsofmeasure.org", "unit": "years"}}
    }
  ],
  "note": [
    {
      "extension": [
        {"url": "http://hl7.org/fhir/uv/genomics-
reporting/StructureDefinition/annotation-code", "valueCodeableConcept":
{"coding": [{"code": "test-disclaimer",
"system": "http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/coded-
annotation-types-cs"}]}}
      ],
      "text": "PRS provides the relative risk of developing a disease and
should be used as an adjunct tool to determine the likelihood of
developing a specific disorder."
    }
  ]
}

```

FIG. 1 Fast Healthcare Interoperability Resource Risk Assessment Example Data instance showing placement of key polygenic risk score data elements.

```

{"resourceType": "Observation",
  "id": "polyGenicScoreObservation",
  "status": "final",
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    "url": "https://paperOnDevelopmentOfScore"}}],
  "category": [{"coding": [{"code": "laboratory", "system": "http://terminology.hl7.org/CodeSystem/observation-category"}]}],
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        "display": "Similarity of Patient's identified genetic ancestry (population) to the score validation population - as a probability (100% is a perfect match)"}]}],

```

FIG. 2 Fast Healthcare Interoperability Resource Observation Example Data instance showing placement of key polygenic risk score data elements.

The examples mentioned earlier demonstrate that when PRS is ready for the clinical audience, FHIR can deliver the data features of PRS computability. FHIR also supports the computation of visual-graphical representations of risk that are easier for clinicians and patients to understand.

The following 2 example flows for FHIR-based PRS reporting provide context for where the example artifacts may be used.

Laboratory reporting flow context

1. Patient physical sample is provided, or patient genetic variation is provided.
 - i. Health features required for calculation are also provided
 - ii. Sequencing performed if required

2. PRSs are calculated using genomic data and demographic data.
 - i. Similarity to study population is evaluated and adjustment is made
3. FHIR Observation instance and RiskAssesment instance similar to the ones exemplified here are produced to be sent.
4. An FHIR DiagnosticReport is packaged up with conclusions and other clinical metadata such as performer and patient identifiers.

Substitutable medical applications, reusable technologies-on-Fast Healthcare Interoperability Resource app flow example context

1. Patient demographic sent as part of CDS Hook

```

      "valueQuantity": {"value": 95, "code": "%", "system":
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validation population"}]}},
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  ],
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"valueCodeableConcept": {"coding": [{"code": "test-disclaimer", "system":
"http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/coded-annotation-
types-cs"}]}], "text": "Percentile is adjusted for patient's similarity to
the study population. Check similarity distance to inform clinical
utility"}]}

```

FIG. 2 (continued)

2. Endpoint for health data queries (eg, FHIR server)
3. Endpoint of server with genomic data (eg, an FHIR server supporting FHIR Genomics Operations [21])
4. Information is sent to the app server
5. On the app server, Polygenomic Risk scores are calculated using the genomic data and demographic data
 - i. Other patient medical context that is needed is pulled from a Health data server.
 - ii. Similarity to the study population is evaluated and adjustment is made.
 - iii. FHIR Observation instance and RiskAssessment instance similar to the ones exemplified

here are produced to be sent to the front-end app.

6. CDS Hooks/SMART on FHIR app returns RiskAssessed Resource with graphical output

Deeper insight into the algorithmic descriptions, such as the population trained on and validation, requires going to the source of the calculation. Definitional FHIR resources such as Measure, ObservationDefinition, and Genomic Study could be explored. However, the FHIR Reference data type has the exquisite ability to provide a URL to make study details accessible.

The final product of the earlier data transmission is a computable artifact supporting calculations, computations, end-user exploration, and communication of clinically critical details.

SUMMARY

HL7 FHIR has tools (specifications) to meet the known and emerging data needs in polygenic risk reporting for clinical decision-making, including supporting representation of continuous risk and ancestry similarity. A question remains regarding the handling of millions of genetic variants that can impact a risk score and finding the balance between delivering care and applying polygenic risk scoring to patients genetically distant from the PRS study population. Currently, the larger corpus of genomic data that fall outside of the few variants disclosed in the clinical report is not available in the clinical environment. How these data are made available to clinicians for generating PRS at the point of care is an open question. The resolution of this question requires engagement with hospitals, laboratories, standards organizations, information technology (IT) specialists, and the relevant regulatory bodies. Trials, such as the GenoVA clinical trial [32], will provide evidence for the critical features required for communication and appropriate application of PRS, but there is work that needs to be done to develop the full stack technical framework for their delivery. The examples here offer a starting point that readily accommodates a technical solution through FHIR's existing tools.

CLINICS CARE POINT

- HL7 FHIR has the tools to enable polygenic risk score reporting.

DISCLOSURES

The authors have nothing to disclose.

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